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Hold the matrix operation fixed.

Compare algorithms for the same four-by-four FP32 inputs.

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C: the same complete 4 × 4 output

1.0	−1.0	0.0	1.0
0.0	−1.0	1.0	0.0
4.0	−1.0	−3.0	4.0
−2.0	4.0	−2.0	−2.0

128 input loads · 16 output stores

Tile width 1 · caches excluded

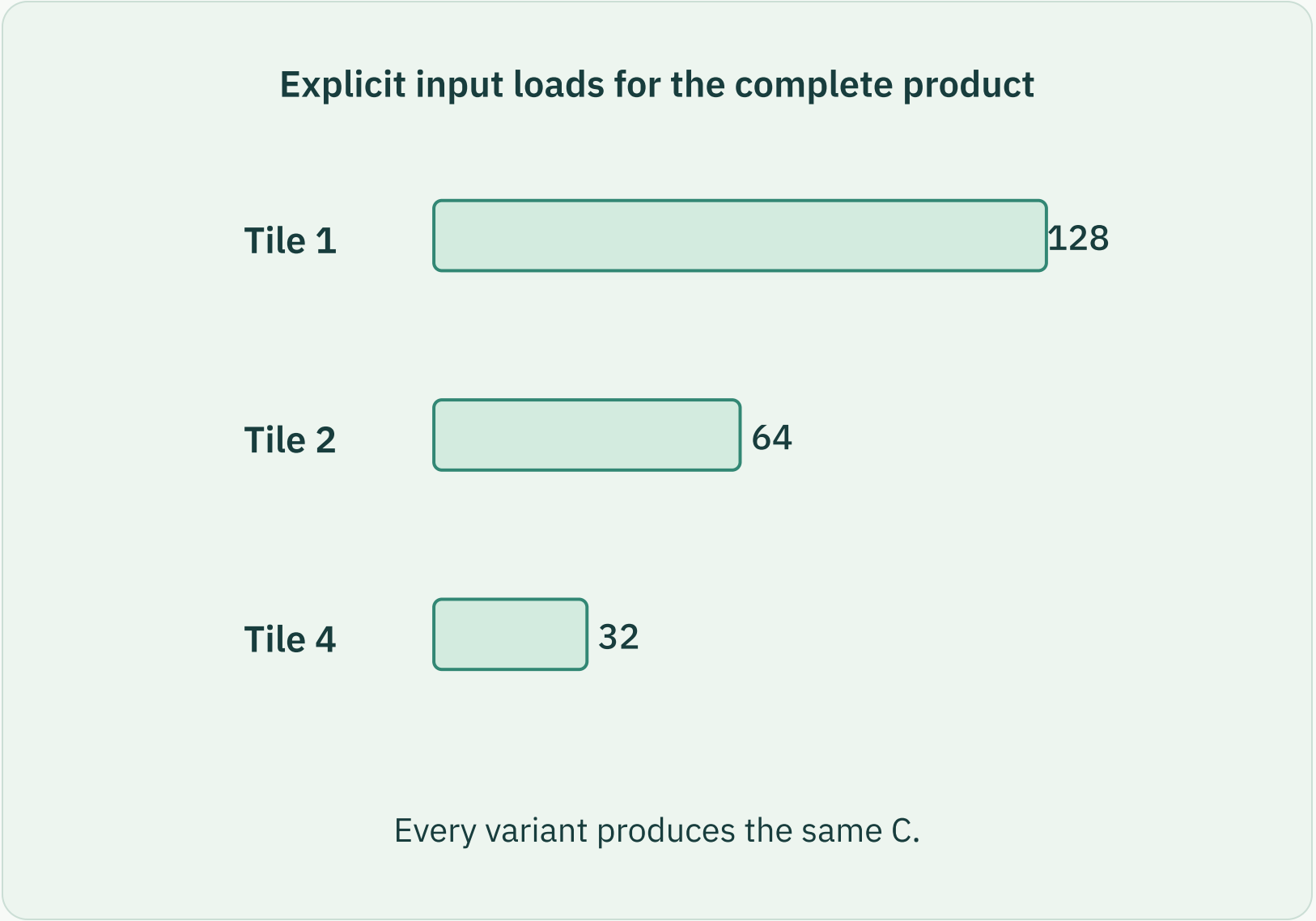
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Share input tiles without changing the result.

Tile widths two and four reduce explicit input loads.

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Larger tiles also consume block resources.

Fewer loads do not establish the fastest implementation.

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Reuse benefit and per-block resource cost

Tile width 2

64 input loads
Complete product

32 shared bytes
Per block

256 registers
Per block

4 blocks fit
Synthetic resource budgets

Tile width 4

32 input loads
Complete product

128 shared bytes
Per block

1024 registers
Per block

1 blocks fit
Synthetic resource budgets

Neither count establishes which kernel is faster.

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Write a hypothesis and its uncertainty.

Choose what to investigate, then name the missing evidence.

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Reuse benefit and per-block resource cost

Tile width 2

64 input loads
Complete product

32 shared bytes
Per block

256 registers
Per block

4 blocks fit
Synthetic resource budgets

Tile width 4

32 input loads
Complete product

128 shared bytes
Per block

1024 registers
Per block

1 blocks fit
Synthetic resource budgets

Neither count establishes which kernel is faster.

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Pause: double the matrix dimension.

Use eight-by-eight inputs and tile width two.

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8 × 8 matrices · tile width 2

Complete-product input loads
?

Output stores
?

Resident-block capacity
?

Does doubling N double every count?

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Different counts grow at different rates.

The per-block resource requirement stays the same.

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8 × 8 matrices · tile width 2

Complete-product input loads
512

Output stores
64

Resident-block capacity
4 blocks in the teaching budgets

Per-block resource requirements are unchanged.

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Measure equivalent implementations fairly.

Check results, define the timed interval, then repeat.

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Design a fair hardware comparison

Check complete outputs
Keep shapes, inputs and dtype comparable.

Define the timed interval
Choose whether copies and allocation are included.

Warm up, synchronise, repeat
Then inspect actual operations with a profiler.

Browser counts are not measured GPU performance.

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Turn the model into a testable recommendation.

Record evidence, assumptions and the next experiment.

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Make a testable recommendation

Evidence
What changed in the browser model?

Uncertainty
Which real implementation costs are missing?

Next measurement
What would support or reject your hypothesis?

Browser counts are not measured GPU performance.

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